Venkata Lakshmi Parimala Pasupuleti

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PROFESSIONAL SUMMARY

AI Engineer with 8+ years of data-intensive production engineering, specializing in **agentic AI systems and LLM-powered analytics**. Built AgentProbe - an open-source multi-agent toolkit (LangGraph, LLM-as-judge eval engine, autonomous agents) and LeaseI - a natural language analytics pipeline over real estate documents (93.6% precision, 82.9% recall). Shipped AI and data pipelines across 1.6M+ financial records and 1M+ property records at Entrata. M.S. Data Analytics Engineering, George Mason University (GPA 3.87, May 2026).

WORK EXPERIENCE

George Mason University

Aug 2024 -- Aug 2025

Research Associate — Data Analytics · Statistical Modeling · Anomaly Detection

- Built regression models ($R^2 = 0.977$) in Python and R across WHO and UK Road Safety datasets spanning 190+ countries; applied Isolation Forest to automate anomaly detection across 3 research pipelines — surfaced anomalies manual review missed.
- Designed Tableau, ggplot2, and Quarto analytics dashboards for faculty leadership — improved research comprehension across two cohorts by 40%; translated complex model outputs into actionable insights for non-technical stakeholders.

Entrata

Jan 2017 -- Aug 2024

Trainee → Associate Engineer → Software Engineer

- Pioneered Entrata's AI engineering foundation — built and shipped the first AI pipeline (Python, PyTorch, OpenAI API) across **1.6M real financial records** with output validation, quality gates, and compliance controls; later productized by the team into a customer-facing feature.
- Built and owned **data pipelines at scale** — 20+ production modules (Python, PostgreSQL, RabbitMQ) across 120+ Agile sprints for 100K+ monthly active users; improved query performance 30% across 200+ SQL queries; 99.5% uptime over 6+ months.
- Developed **affordable housing compliance automation** (LIHTC/Tax Credit, HUD, Section 8/HCV) — income eligibility verification, AMI threshold calculations, recertification workflows across 100K+ unit portfolios; eliminated 80% of manual operational workload via Python/PL-SQL automation.
- Achieved **100% PHI compliance** on AWS S3 migration (Libsodium-encrypted pipelines, zero security incidents); cut execution time 50% via Docker containerization; led 300+ peer code reviews.

PROJECTS

AgentProbe: Multi-Agent AI Testing & Evaluation Toolkit

Open Source

Python · LangGraph · LangChain · FastAPI · ChromaDB · OpenAI · Anthropic · React · Docker · GitHub Actions — github.com/venkatalakshmiparimala/AgentProbe

- Built a **3-agent LangGraph runtime** (Orchestrator, RAG Tracer, Auto Fix) with tool calling, inter-agent state handoffs, and multi-provider LLM routing — autonomous agents that identify issues, trace retrieval quality, and self-correct without human intervention.
- Implemented **5-metric LLM-as-judge eval engine** (factual accuracy, hallucination, retrieval, completeness, compliance) with CI/CD quality gates; React dashboard tracking per-run metrics, regression drift, and latency across concurrent agent test suites.

LeaseI: Natural Language Analytics over Real Estate Documents

Python · LangGraph · SQLite · reportlab — github.com/venkatalakshmiparimala/Lease_complaince_rag

- Built an agentic LangGraph pipeline enabling natural language Q&A over 1,240 residential lease documents for compliance gap detection across 10 rules, 5 US states + federal standards — **93.6% precision, 82.9% recall, 87.9% F1**; deterministic rules engine keeps decisions fully auditable.
- Implemented pass@k eval harness (pass@1 81.7%, pass@3 88.3%) with regression gate; LLM-as-judge scores both accuracy and reasoning quality — analytics-to-action pipeline surfacing compliance gaps and recommending remediation steps.

CrashClarity: Large-Scale Accident Analytics (104K Records)

Python · R · Scikit-learn · Random Forest · Logistic Regression · Feature Engineering · GitHub Pages — Live Dashboard

- Achieved **85%+ accuracy** on 104K+ UK accident records — Random Forest + Logistic Regression pipeline with feature engineering across weather, road, and time variables; deployed interactive analytics dashboard surfacing actionable safety insights on GitHub Pages.

EDUCATION

M.S. Data Analytics Engineering — George Mason University — Graduated May 2026

GPA: 3.87 · Fairfax, Virginia

B.Tech Computer Science & Engineering — Sree Venkateshwara College of Engineering — 2013 -- 2017

TECHNICAL SKILLS

Languages: Python, SQL, PL/SQL, R, JavaScript, TypeScript, Bash, HTML/CSS

AI & Agents: LangGraph, LangChain, Agentic AI, Multi-Agent Orchestration, RAG Pipelines, LLM-as-Judge, LLM Evaluation, Prompt Engineering, ChromaDB, OpenAI API, Anthropic API, Embeddings, Semantic Search, NLP, Agent Observability, FastAPI, REST APIs

Analytics & ML: PyTorch, Scikit-learn, XGBoost, Random Forest, Isolation Forest, Anomaly Detection, Feature Engineering, Statistical Modeling, Regression Analysis, NumPy, Pandas, Tableau, Power BI, ggplot2, Jupyter

Data & Infrastructure: PostgreSQL, MySQL, AWS (S3, EC2, RDS, Glue), Docker, CI/CD (GitHub Actions), ETL, Data Pipelines, Microservices, RabbitMQ, Git, Linux, Agile/Scrum